A3-PS015-TC041

Explainable Home and Last-Mile Connectivity Diagnosis (PS-015, Smart Cities)

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AIORI Hackathon 3

October 18, 2026

Problem Description

- Speed tests report peak download bandwidth on an unloaded link and hide packet loss, round-trip latency and loaded jitter, which decide real-time application quality.
- Users see "internet is slow" but cannot tell a local gateway fault from DNS trouble, packet loss or congestion.
- Access media (fiber, cellular, fixed wireless, satellite) behave differently, so one threshold does not fit all.
- Challenge: a guided desktop diagnostic that explains each finding with test evidence and produces an escalation report, or says "Inconclusive", when the cause cannot be localised.

Background

- Key metrics: reachability, round-trip latency, jitter, packet loss, DNS resolution delay and throughput [1]–[4].
- Last mile: the home LAN/Wi-Fi, the gateway and the access link to the ISP. Evidence from a single endpoint can rarely prove where on this path a fault lies.
- Loaded latency (latency while the link is busy) exposes queueing and congestion that idle tests miss.
- Limits: ICMP may be filtered, Wi-Fi interference cannot be claimed without radio telemetry, and no damaged-cable or ISP-internal root-cause claims are made.

Literature & RFC Review

- RFC 2330 [1]: framework for IP performance metrics (repeatable, well-defined measurement, singletons and samples). Our probe design follows it.
- RFC 2681 [2] and RFC 2680 [3]: round-trip delay and one-way packet loss metrics. We apply the loss definition and report sent, received and lost probes.
- RFC 3393 [4]: IP packet delay variation, the basis for our jitter calculation.
- RFC 6349 [5]: TCP throughput testing, guiding idle versus loaded throughput measurement.
- RFC 1035 [6] and RFC 792 [7]: DNS and ICMP behaviour behind the resolver and gateway probes.

Proposed Solution

- Python desktop diagnostic agent with a guided workflow: gateway, DNS, latency/loss/jitter, then idle vs loaded throughput.
- Transparent rule engine. Each rule reads named measurements, compares them to a healthy baseline and returns a finding with its evidence. Several findings can co-exist.
- Rules: gateway failing repeatedly; DNS failing while a direct IP works; loss above baseline; loaded latency and jitter well above idle; otherwise Inconclusive.
- Controlled lab (network namespaces or isolated VMs) with fault-injection and reset scripts for the four faults.
- Escalation report (PDF) with timestamps, raw samples, diagnosis, limitations and next steps.

Optimization Proposed

Aspect	Before (standard speed test)	After (this tool)
Metrics	Peak download speed	Reachability, DNS delay, latency, loss, jitter, throughput
Load condition	Unloaded only	Idle and loaded compared
Diagnosis	None; single number	Rule-based with cited evidence
Uncertain cases	Not handled	Returns Inconclusive
Escalation	Manual screenshots	Evidence-based report export

Solution Architecture & Design

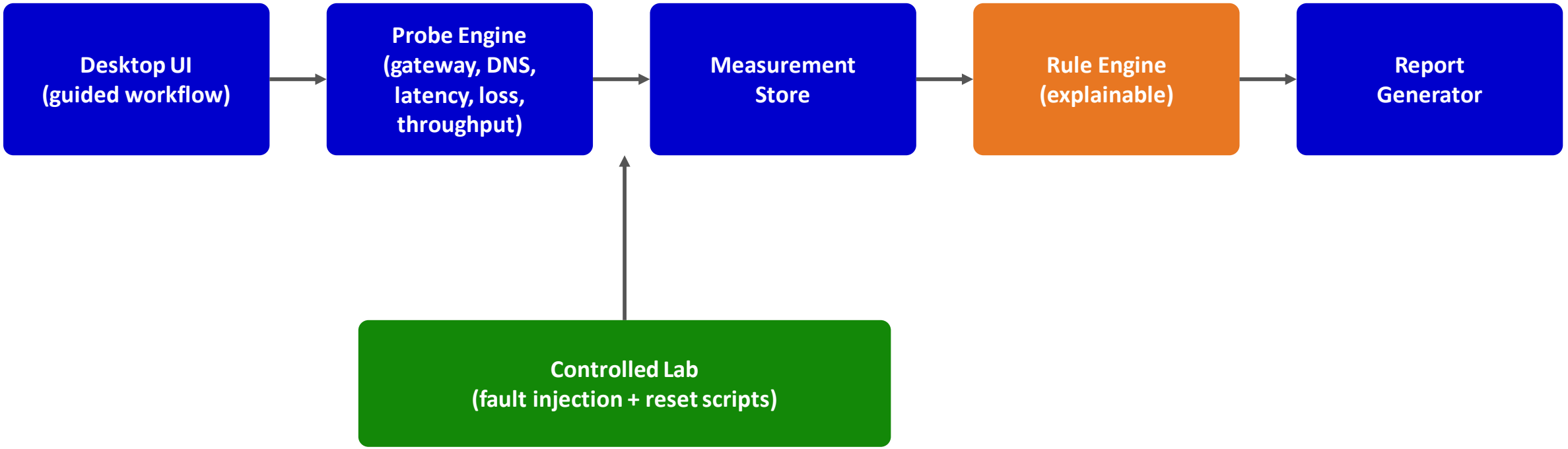


Figure: System architecture (agent → measurements → rule engine → report)

Data Flow Diagram

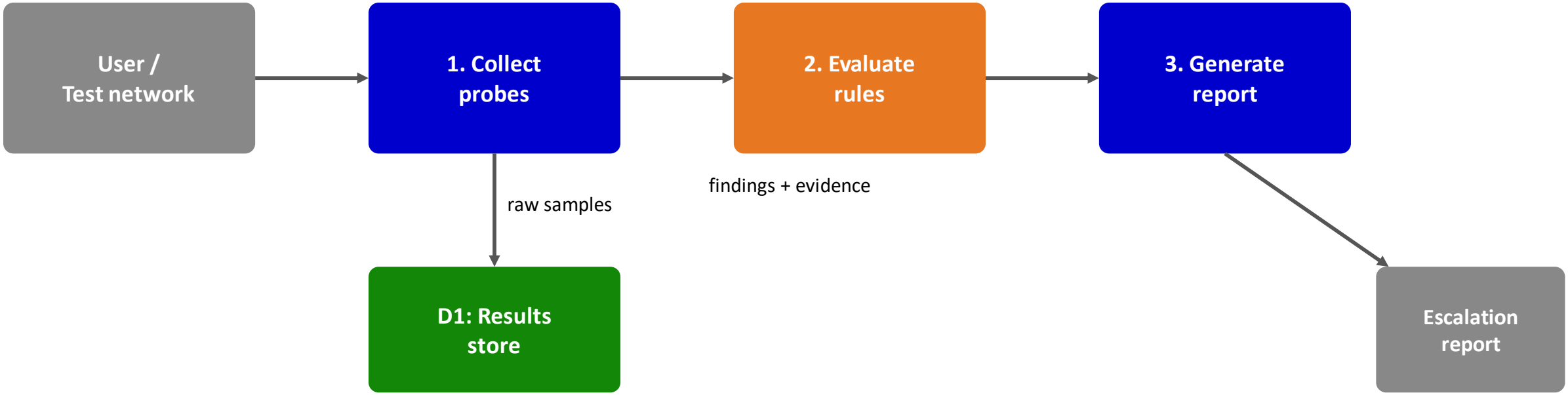


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Phase	Milestone	Output
1. Foundation	Set up repo, Python env, gateway and DNS tests, basic dashboard	Working prototype
2. Detection	Loss, jitter and loaded throughput; rule engine; fault scripts	Four faults detectable
3. Evidence	Validate all faults, document an inconclusive case, build report	Evaluation + PDF report
4. Delivery	Repo hand-over to aiori-hackathon, mentor demo with pseudocode	Final demonstration

Proposed internal phases, not official hackathon deadlines.

Bibliography

[1] V. Paxson et al., "Framework for IP Performance Metrics," RFC 2330, IETF, 1998.

[2] G. Almes, S. Kalidindi, M. Zekauskas, "A Round-trip Delay Metric for IPPM," RFC 2681, 1999.

[3] G. Almes, S. Kalidindi, M. Zekauskas, "A One-way Packet Loss Metric for IPPM," RFC 2680, 1999.

[4] C. Demichelis, P. Chimento, "IP Packet Delay Variation Metric for IPPM," RFC 3393, 2002.

[5] B. Constantine et al., "Framework for TCP Throughput Testing," RFC 6349, 2011.

[6] P. Mockapetris, "Domain Names - Implementation and Specification," RFC 1035, 1987.

[7] J. Postel, "Internet Control Message Protocol," RFC 792, 1981.

Thank You for Your Attention!